6. Potential Drivers of Disparate Privacy

The evidence reviewed in the preceding sections shows that privacy exposure can vary substantially across records, patients, and clinical populations. This section examines five candidate explanations for that heterogeneity: selective memorization and generalization behavior; rarity, representation, and distributional differences; model capacity and training procedures; the multiplicity and dependence of patient records; and sample or patient influence. These explanations are related but non-equivalent. The available literature therefore supports treating them as interacting hypotheses whose independent and causal contributions remain to be established.

6.1 Memorization and Generalization

Memorization, overfitting, generalization gap, and membership vulnerability are related but non-equivalent. Memorization is example-specific dependence: including a training example changes what the model predicts or assigns high likelihood to beyond what can be inferred from the remaining data. In supervised learning, it can be measured by comparing predictions when an example is included versus excluded; in generative models, it may appear as unusually high exposure of a rare sequence. Overfitting denotes adaptation to training data that impairs performance on unseen data, whereas the generalization gap is an aggregate difference between training and test or population loss. Membership vulnerability is adversarial: it is the ability to distinguish inclusion from exclusion under a specified attack, prior, and access model. A model may therefore memorize selected examples without a large average generalization gap, and memorization does not guarantee successful membership inference.

Yeom et al. provide the clearest formal connection between overfitting and membership inference [10]. For their bounded-loss threshold attack, membership advantage is related to average generalization error; under the conditions of that attack, a positive generalization gap is sufficient to create an exploitable membership signal. Their experiments also associate increasing overfitting with greater attack success. However, this relationship concerns a particular attack and an expectation over training sets and examples, not the vulnerability of every record. Yeom et al. further show that overfitting is not necessary for privacy leakage in general: a learning algorithm can reveal membership while retaining good predictive performance or stability. Consequently, an aggregate association between generalization gap and MIA success neither proves that every vulnerable record was memorized nor guarantees that a small gap eliminates leakage.

Strong low-FPR attacks can succeed against well-generalized models, showing that aggregate overfitting is insufficient to explain every exposed example [2]. Sample-specific separability should be assessed against comparable non-members rather than inferred from a population generalization gap.

Sample-level work offers a more precise account of selective dependence. Feldman's long-tail model shows that memorization may be statistically useful, rather than simply erroneous, when data contain many low-frequency subpopulations: fitting rare but informative examples can be necessary for close-to-optimal generalization [27]. Feldman and Zhang operationalize this idea using memorization and influence estimates [28]. Memorization compares the probability of correctly predicting an example when it is included versus excluded, whereas influence measures how that inclusion changes performance on other examples. On non-clinical benchmarks, highly memorized samples included atypical observations, outliers, and mislabeled examples, but removing memorized examples could also reduce test accuracy. Song and Mittal likewise found heterogeneous sample-level privacy-risk scores associated with model sensitivity, class-level generalization, and representation properties [15]. These findings make example-specific dependence a plausible link to membership vulnerability, but they do not make privacy vulnerability, influence, and memorization interchangeable measures.

The clinical tail [3] is consistent with selective memorization but does not prove it. Rarity, subgroup representation, record count, label difficulty and architecture co-vary; record-specific IN/OUT separability establishes vulnerability, not storage of a recoverable patient record.

6.2 Data Rarity, Representation, and Distributional Differences

Rarity arises at several levels that should not be treated as equivalent. Population-group rarity describes a demographic or clinical subgroup contributing relatively few patients or records; diagnosis or class rarity concerns a low-prevalence label; and local record rarity describes an observation with few close analogues in the relevant feature space, even within a large group. Distributional difference is distinct again, arising from institution, device, acquisition protocol, preprocessing, or clinical workflow. Long-tail theory suggests that learning rare subpopulations may require example-specific memorization [27], while influence estimates show that highly memorized examples often include atypical observations [28]. Yet group size is not a proxy for local density: a patient from a small subgroup may have many clinically similar neighbors, whereas an unusual patient may belong to a majority group. These forms of rarity must therefore be measured separately.

The strongest direct clinical evidence remains observational. Knolle et al. found that some underrepresented disease, demographic, insurance, and imaging-protocol categories appeared disproportionately in the highest record-level MIA-AUC percentile [3]. This supports an association between representation and extreme exposure but does not isolate low representation as its cause; disease difficulty, label composition, record multiplicity, acquisition context, and model configuration may also shape the tail. Wang et al. similarly reported stronger subject-level participation leakage in small or institution-specific ECG cohorts, whereas larger and more diverse datasets showed lower subject-level attack vulnerability [22]. However, their cross-dataset non-members originated from different datasets, allowing attacks to exploit institutional, device, resampling, or preprocessing signatures. These findings demonstrate cohort-associated leakage but do not cleanly separate within-institution membership from dataset-origin recognition.

Controlled changes to class representation offer evidence closer to causal intervention, although not at clinical patient level. Truex et al. found greater membership-inference vulnerability for minority classes in skewed datasets, with attack vulnerability also varying with overfitting [30]. Hu et al. constructed long-tailed CIFAR datasets with controlled imbalance factors and observed increasing low-FPR attack success from head toward tail classes [31]. Tail vulnerability persisted even when long-tailed learning improved tail accuracy and differential privacy reduced overall leakage. These experiments strengthen the claim that changing representation can change class-conditional exposure. Nevertheless, they concern benchmark classes rather than patients, and representation changes also alter optimization, loss, accuracy, and effective overfitting. They do not establish that reducing a clinical subgroup's prevalence will invariably increase every member's vulnerability.

Local rarity shifts attention from group counts to a target's location within the distribution. Long et al. showed that well-generalized models can contain vulnerable records whose inclusion produces distinctive influence not captured by aggregate overfitting; although their experiments included a cancer benchmark, they did not constitute a modern patient-level clinical audit [32]. In a simplified fixed-target game for an empirical-mean mechanism, Azize and Basu characterized optimal leakage as increasing with the target's Mahalanobis distance from the data-generating distribution [33]. DOMIAS instead measures local overfitting in generative models by comparing synthetic-model and reference densities and was particularly effective against uncommon samples [34]. These results support a plausible mechanism: when few comparable observations surround a target, its inclusion may change model behavior more distinctly. However, theoretical distance and generative density ratios are not direct measures of vulnerability in clinical prediction models.

Distributional differences and reference mismatch further complicate interpretation. Humphries et al. showed that dependencies between member and non-member sources, including hospital-specific distributions, can increase measured MIA success relative to an IID evaluation [23]. Shi et al. found no uniform direction: cutout, jitter, Gaussian noise, label shift, and attribute shift could increase or decrease attack performance, sometimes oppositely across attacks [25]. Watson et al. additionally showed that uncalibrated scores can confuse membership signal with intrinsic sample difficulty; low loss may simply indicate that an example is easy even for models trained without it [13]. Audits should therefore use patient-disjoint splits, within-site comparators, site- or protocol-matched reference models, and difficulty calibration. Sensitivity analyses should also condition on diagnosis, acquisition variables, record multiplicity, and the definition of a clinically meaningful neighborhood.

Rarity is a family of hypotheses, not a patient label. Matched analyses must separate group size,

clinical prevalence and local density from difficulty, source institution, record count and label noise. The causal question is whether changing one of these factors changes the same patient's membership vulnerability [13,27,28].

6.3 Model Capacity and Training Procedures

Clinical model configurations with greater capacity had larger extreme-vulnerability tails [3]. Parameter count was not isolated: architecture, resolution and pretraining differed across configurations. The result is therefore an association with a configuration, not a causal parameter-count effect.

The broader literature likewise provides no basis for assuming a universally monotonic relationship between capacity and privacy. Tan et al. theoretically showed, in an overparameterized linear-regression setting with Gaussian data, that membership-inference vulnerability can increase with the number of model parameters; their experiments in several non-clinical learning settings supported a similar pattern [36]. This provides a plausible explanation for why highly expressive models may encode stronger signals about particular training examples. A complementary analysis by Tan et al. demonstrated that the direction of the relationship can change when regularization is coordinated with capacity [37]. In their theoretical logistic-regression setting, and empirically in neural networks trained with early stopping, larger models could achieve both improved generalization and stronger resistance to membership inference than smaller alternatives. Thus, capacity cannot be evaluated independently of the procedure used to constrain or terminate learning. These results are not direct evidence about clinical patients, but they prevent the clinical association reported by Knolle et al. from being interpreted as a universal scaling law. Capacity is a context-dependent vulnerability modifier rather than an independent or monotonic cause of privacy leakage.

More training can fit member-specific details, but regularization, early stopping and capacity interact. General-benchmark results show that the effect of overparameterization can reverse under appropriate regularization [36,37]. A clinical causal comparison should hold data, architecture and utility targets fixed while varying one training decision at a time.

Data augmentation further illustrates why training procedures cannot be assigned a fixed privacy effect. Kaya and Dumitras found that augmentation intensities selected primarily to maximize predictive utility did not substantially mitigate membership inference across their non-clinical image-classification experiments [38]. Stronger augmentation sometimes reduced attack success, but this protection was accompanied by a utility penalty. The authors also found that label smoothing could amplify membership vulnerability, demonstrating that procedures commonly classified as regularizers do not necessarily protect privacy. Moreover, the privacy-utility behavior of high-intensity augmentation resembled that of reducing the number of training epochs: both weakened the membership signal while limiting model fitting. Augmentation may therefore alter privacy through several pathways, including effective training duration, invariance, and example-specific difficulty. Because these results were obtained from general image benchmarks, they motivate clinical sensitivity analyses but do not establish that the same effects will occur for medical images, physiological signals, or longitudinal records.

Capacity can raise vulnerability broadly, expand the upper tail or change which patients are exposed. The clinical configuration comparison above confounds architecture, resolution and pretraining. Factorial comparisons should isolate these choices; tail and paired patient results follow Table 4.1.

6.4 Record Multiplicity, Correlation, and Patient Contribution

Record multiplicity refers to the number of records contributed by patient p, denoted by m_p, whereas contribution weight describes how strongly that patient enters the training objective through record count, sampling frequency, or repeated exposure across epochs. Under record-uniform empirical-risk minimization, the patient's nominal weight is proportional to the fraction of all training records that the patient contributes:

$$w_p = \frac{m_p}{\sum_q m_q}.$$

Patient-uniform sampling can remove this mechanical weighting without eliminating the informational effects of repeated records. A third concept, within-patient correlation, captures dependence among images, visits, or physiological signals belonging to the same individual. Patients with equal record counts may therefore contribute very different information: ten nearly identical images do not have the same effective contribution as ten clinically and temporally diverse visits. Multiplicity may affect privacy through three mechanisms. An opportunity effect occurs when patient vulnerability is defined by the maximum record score, because patients with more records have more opportunities to produce an extreme value. A training-influence effect may arise when repeated records give one patient greater weight in the objective or gradient updates. A shared-signature effect may occur when correlated records reinforce stable subject-specific features, including anatomy, chronic disease patterns, or acquisition characteristics. The first mechanism may be partly a measurement artifact, whereas the latter two represent potential changes in what the model learns.

Knolle et al. provide the principal clinical evidence that multiple-record contributions change the appropriate audit [3]. Their target subsets were sampled at patient level, so all records from a patient were included or excluded together; preliminary record-level sampling underestimated attack success when related records remained in training. The study summarized patient vulnerability using the maximum record-level MIA AUC and also showed why record-level differential privacy does not automatically protect a complete patient contribution. However, it did not vary (m_p) independently. A maximum can rise mechanically with record count, while record count may also reflect disease severity, protocol, or follow-up duration. The study therefore establishes the importance of patient-level construction, not a causal dose-response relationship between record count and vulnerability.

Within-patient records also violate the independence assumptions underlying many privacy audits. Repeated observations may share anatomy, biological characteristics, chronic diagnoses, institutional equipment, acquisition protocols, preprocessing pipelines, or temporal context. Humphries et al. showed that statistical dependencies between training examples can substantially increase membership-inference success and that evaluations assuming independent and identically distributed records may underestimate leakage [23]. Their experiments do not directly reproduce longitudinal records belonging to individual clinical patients, so they provide general mechanistic support rather than a clinical estimate of the effect's magnitude. The methodological implications are nonetheless important: training, test, and reference sets should be separated at the patient level; reference-model datasets should preserve realistic dependency structures; and record scores from the same patient should not be treated as independent probabilities. Statistical uncertainty should likewise be estimated through patient-level resampling or cluster bootstrap procedures rather than by independently resampling correlated records.

Subject-level attacks show that exposure can arise from the joint structure of a person's records. Suri et al. infer whether a subject's distribution contributed to federated learning without requiring the exact training records [16]. Li et al. use compactness among a user's embeddings and observed higher recall for users with more training images [17]. Both studies are non-clinical, and the latter concerns person re-identification, so they provide mechanistic support rather than clinical effect estimates. They nevertheless clarify the need to separate records used for training from records available to the attacker:

$$m_p^{\text{train}} = \text{number of patient records used for training},$$

$$q_p^{\text{attack}} = \text{number of patient records available to the attacker.}$$

Increasing the former may change what the model learns about the patient, whereas increasing the latter may improve only the attacker's statistical power. Their effects should not be conflated.

A monotonic relationship between record count and privacy vulnerability should not be expected. Near-duplicate records may increase a patient's effective training weight, whereas diverse records may improve representation of the patient's underlying distribution and reduce dependence on any single observation. Additional records could consequently reduce record-level identifiability while increasing subject-level distinguishability. Highly correlated records also contribute less independent information than an equal number of heterogeneous records, and a maximum-based patient score

can rise with m_p even if the model's treatment of every record remains unchanged. Results from clinical auditing, dependent-data experiments, and subject-level attacks therefore support several competing mechanisms rather than a universal direction of effect [3,16,17,23]. Record multiplicity is a configuration-dependent vulnerability modifier whose effect depends on within-patient correlation, sampling weight, the membership hypothesis, the attacker's auxiliary records, and the patient-level decision rule.

Controlled clinical subsampling should vary training record count separately from attacker-held records and within-patient similarity, with diagnosis, institution and cohort size controlled. Patient-level splitting is necessary, but the independent effect of multiplicity remains unknown.

6.5 Sample Influence and Related Explanatory Measures

Data rarity and record multiplicity describe properties of a patient's contribution, but neither establishes that the contribution materially changed the trained model. A deletion effect addresses this counterfactual more directly. For a record z_i and evaluation quantity T, define

$$I_i^{\text{del}}(T) = \mathbb{E}_\xi \left[T(f_{D,\xi}) - T(f_{D\setminus\{z_i\},\xi}) \right],$$

where ξ represents training randomness. For patient p, who contributes record set $\mathcal{R}_p$, the corresponding quantity is

$$I_p^{\text{del}}(T) = \mathbb{E}_\xi \left[T(f_{D,\xi}) - T(f_{D\setminus\mathcal{R}_p,\xi}) \right].$$

The sign depends on the direction of T and should be declared; the magnitude alone should be used only when direction is intentionally ignored. Interpretation also depends on T: self-influence concerns change in the patient's own loss or prediction; cross-influence concerns effects on other examples; broader quantities may assess overall or subgroup clinical utility; and attack-oriented influence concerns changes in observable statistics such as confidence or loss. These deletion effects are distinct from influence-function approximations, memorization, membership vulnerability, and intrinsic difficulty. A patient may strongly influence clinical utility without being easily identifiable, or minimally affect aggregate utility while producing a distinctive membership signal.

Membership vulnerability concerns separation of full IN and OUT distributions of an attacker-observable statistic, not simply a mean deletion effect. Nonclinical studies connect influential examples to attacks even under good generalization [32] and associate sample-specific posterior scores with sensitivity and representation [15]. Knolle et al.'s resampling result is clinical record-specific separability, but not paired causal deletion [3]. Influence remains a candidate mediator, not a validated privacy proxy.

Explicit leave-one-out retraining provides the most direct operational estimate of a deletion effect: train a model on D, retrain it after removing the target contribution, keep the remaining data and protocol fixed as closely as possible, and compare losses, predictions, attack statistics, and utility across repeated seeds. Feldman and Zhang developed repeated-subsampling estimators of memorization and influence that make related inclusion-exclusion comparisons more scalable [28]. Their results also showed that highly memorized examples can improve predictions for rare test examples, indicating that elevated influence is not necessarily undesirable. In clinical auditing, the complete patient contribution must be removed,

$$D_{-p} = D \setminus \mathcal{R}_p,$$

rather than deleting one record while retaining correlated observations from the same person. Patient influence also cannot generally be assumed to equal the sum of record influences because correlated records may interact nonlinearly:

$$I_p^{\mathrm{del}}(T) \neq \sum_{j=1}^{m_p} I_{pj}^{\mathrm{del}}(T).$$

Direct retraining remains expensive and sensitive to initialization, optimization, augmentation, and changes in the fitted solution. Repeated, preferably paired training designs are consequently required to distinguish systematic patient influence from ordinary training variability.

Influence functions offer a cheaper approximation when exhaustive retraining is infeasible. Koh and Liang estimate how infinitesimally upweighting an example changes parameters and a selected loss or prediction [39]. This traces an output to training data but is not itself a privacy metric, and its classical assumptions do not hold exactly for deep non-convex networks. Patient-level deletion is also a grouped, non-infinitesimal perturbation. Koh et al. found that group-influence estimates could rank removal effects usefully while retaining substantial absolute and relative error [40]. Approximate rankings may therefore screen patients for detailed study, but require validation through actual leave-one-patient-out retraining before they support causal or privacy claims.

Candidate measures include loss, confidence, gradient similarity, sensitivity, memorization, representation distance, local density, leverage, and cross-run disagreement. Each captures only part of the mechanism. Low loss may indicate intrinsic ease rather than leakage [13]; large gradients may reflect label noise; and representation distance may measure rarity without proving an inclusion effect. Influence on utility likewise need not match influence on an attack statistic. Because exposure is relative to the surrounding data—as illustrated by the Privacy Onion Effect [35]—these variables should be analyzed separately and conditionally rather than collapsed into an unvalidated composite score.

A candidate influence measure should be validated against repeated leave-one-patient-out retraining for a representative sample, then related separately to attack statistics and clinical utility. Reducing measured influence is evidence of a mechanism only if the same patient's vulnerability falls without compensating harm or exposure elsewhere.

6.6 Strength of Evidence and Unresolved Causal Questions

Association, intervention and formal explanation support different claims. The direct clinical work synthesized in Sections 5 and 6 establishes heterogeneous exposure but does not isolate its drivers. Table 6.1 separates clinical observation from mechanistic support; tiers in Table 3.2 describe transfer to the target clinical claim.

Table 6.1 summarizes the resulting evidential position. The categories are descriptive syntheses rather than a formal GRADE-style rating and should be interpreted under the appraisal approach defined in Section 2.4.

Proposed driver	Direct clinical evidence	Controlled or mechanistic support	Current causal conclusion
Memorization and generalization	Clinical patterns are consistent, but the mechanism was not directly measured	Relatively strong	Plausible mechanism without patient-level causal validation
Rarity and representation	Record-level clinical associations	Controlled class-imbalance experiments	Potential amplifier with substantial residual confounding
Capacity and training	Association with model configuration	Mixed results across training regimes	Conditional modifier rather than a monotonic cause
Multiplicity and correlation	Strong evidence for the correct audit unit	Dependent-data and subject-level evidence	Independent dose-response effect remains unresolved

Proposed driver	Direct clinical evidence	Controlled or mechanistic support	Current causal conclusion
Sample influence	No direct clinical validation	Theoretical, subsampling, and approximation evidence	Candidate mediator not validated for patient-vulnerability identification

Causal tests should ask whether rarity matters after difficulty matching, whether record count matters at fixed correlation, whether architecture or parameter count expands the tail, and whether selective influence reduction lowers exposure without utility loss or transfer [35]. These are hypotheses for Section 9, not established clinical mechanisms.